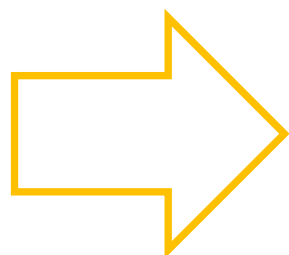




Multi Axis and Dual Axis Motion Controllers



User Manual

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[original document]



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Preface

Products Covered

This manual covers the following Veeco products:

- Multi-Axis Motion Controller (MAC Controller)
- Dual-Axis Motion Controller (DMC Controller)

If you have a different Veeco product, please contact us using the [Technical Support](#) information on page 3 to receive the correct manual.

Intended Audience

This manual is intended for trained molecular beam epitaxy (MBE) professionals, technicians, and persons under the direct guidance of such individuals. If you are not among these groups, please do not attempt to use the information in this manual to operate the equipment.

Using This Manual

To make best use of this manual:

- **Read the entire manual first.** Do not attempt to operate or perform maintenance of any kind on the product before you have thoroughly reviewed this manual.
- **Pay close attention to all safety information!** All MBE applications include inherent hazards and require strict adherence to safety standards. Read [Operator Safety](#) on page 5 for critical safety information. Also see [Safety Hazards](#) on page 2 to learn how safety hazards are indicated in this manual.
- **Use this manual as a tool for putting your own knowledge into practice.** This manual does not cover the theory, principles, or best practices for any particular MBE application. It aims to provide useful information to help you achieve your own objectives.
- **Remember that many Veeco products are configurable.** It is not possible to address all aspects of all configurations in a single manual. If you do not find the information you need, please consult the [Additional Documentation](#) listed on page 2, or contact us using the [Technical Support](#) information on page 3.

- **Refer to all graphics in context.** The graphics in this manual may not exactly match your product. Graphics are intended to illustrate only the features relevant to the topic at hand. Any optional, configurable, or missing features are identified, if contextually relevant.

Document Conventions

Safety Hazards

The following formats are used to communicate safety hazards. Where applicable, additional hazard-specific symbols may accompany these formats.



WARNING — HAZARD SUMMARY. Indicates a hazard that, if not avoided, may result in injury or death to personnel, or long-term adverse effects on health.



CAUTION — Indicates a hazard that, if not avoided, may result in damage or destruction of equipment.

Notes, Tips, and Important Messages

The following formats are used for various types of information that do not directly pertain to hazards.

NOTE — Indicates additional explanatory information.

TIP — Indicates a suggestion to help you perform a task successfully or understand something better.



IMPORTANT — Indicates essential information that must be considered or acted upon.

Additional Documentation

Several other Veeco documents pertain to the Multi Axis and Dual Axis Motion Controllers. Please refer to these documents as needed:

- MBE system manual
- Veeco Valved Cracker or Valved Source user manual

For components not manufactured by Veeco (pumps, gauges, valves, etc.), always consult the original equipment manufacturer (OEM) manual.

Technical Support

For technical assistance or questions, please use the contact information below.

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Hours: Monday – Friday, 8 a.m. – 5 p.m. CST

Documentation Feedback

Veeco strives to continuously improve its technical publications. If you find errors, omissions, or inconsistencies in this manual, please contact Veeco Technical Support and describe the problem.

Your feedback is appreciated.

Version History

Revision	Date	Description of Changes
A	11/16	Initial release.
B	3/17	Added contents for customizing axis names, data logging, flipper control.
C	7/17	Updated Molly software instructions.
D	12/17	Added motor torque table and updated contents for software changes.
E	10/18	Added motor caution and overall document updates.

Notices

Veeco has no control over how the product will be used for research or production purposes and is not responsible for personal injury or damage resulting from such use. It is the sole responsibility of users of the product to comply with all local, state, and federal safety laws and regulations applicable to the system.

The content of this publication is subject to change without notice. Veeco Instruments, Inc. reserves the right to make product-improvement changes which may or may not be reflected in this publication. Veeco Instruments, Inc. is not responsible for any inadvertent omissions or errors. Veeco Instruments, Inc. assumes no liability for damages arising out of, or in connection with, the application or use of any product or application described herein.

This manual does not describe or constitute any commitment, agreement or relationship between Veeco Instruments, Inc. and the customer. Please refer to the Sales Contract for warranty terms and obligations.

If the customer alters the product, the customer assumes responsibility for safety, installation, and operation issues, and all other items pertinent to conformance to standards.

Veeco accepts no responsibility for failure to follow the safety information included in this manual, or failure to comply with all applicable health, facility, and safety guidelines.

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Chapter 1 Important Information

Operator Safety

Safety Overview

The MAC and DMC Controllers have been designed and manufactured with strict regard for the safety of operators. However, as with any sophisticated instrument, safety also depends on proper use and maintenance throughout the life of the product.

Carefully study this manual and all safety information in the manual for your MBE system. For additional information on operator safety, refer to the appropriate health, safety, and facility guidelines.



IMPORTANT — Make sure that anyone who operates or maintains the MAC or DMC Motion Controller is properly trained and fully understands how to reduce or eliminate safety hazards.

Warnings



WARNING — READ THIS MANUAL. Read this entire manual before attempting to install, operate or maintain this product. Failure to read this entire user manual and follow all safety information contained herein could result in serious injury or death.



WARNING — HIGH VOLTAGE. Make sure that ALL sources of input power to this product are turned OFF and disconnected before installation or service. Failure to do so could result in serious injury or death.



WARNING — DO NOT MODIFY. This product has been designed to function safely when used as described in this manual for standard applications. If it is modified for any nonstandard application or material selection, hazardous conditions may result.



WARNING — HOT SURFACES. Surfaces on the servo motor could become hot during operation. If the servo motor is in an exposed application, use care to avoid contact.

Hazard Alert Labels on Equipment

Hazard alert labels are applied to various locations on the MAC and DMC Controllers.

The table below shows labels that may appear on the product and describes how to interpret them.

Hazard Alert Label	Hazard Indicated	Explanation
	High voltage.	See HIGH VOLTAGE warning on page 5.
	Hot surfaces	See HOT SURFACES warning on page 5.

Product Cautions



CAUTION — This product may be damaged or destroyed if installed, configured, operated, or serviced improperly. Follow the product specifications and observe all specific cautions included in this manual. Installation, operation, and service should be performed by trained technicians only.



CAUTION — Parts of this product are fragile and may be damaged if handled improperly. Always ship the product using the original packaging materials if possible.



CAUTION — Damage could occur if the wrong servo motor is accidentally installed onto a component. The hazard is greatest for axes of motion configured for automated valve positioning. To avoid damage, ensure that the original servo motor gets re-installed onto each component whenever multiple servo motors have been removed for system bake or maintenance.

Product Specifications

Item	Specification
Mechanical	
Chassis type	MAC 5U Full Rack DMC 2U Full Rack
Physical dimensions	MAC 17.16 in (436 mm) wide 8.72 in (221.5mm) high 11.5 in (292 mm) deep DMC 17.16 in (436 mm) wide 3.54 in (90 mm) high 9.64 in (245 mm) deep
Weight	MAC 16 lb. (7.3 kg) approx. DMC 5 lb. (2.5kg) approx...
Electrical	
Input voltage	110–230 VAC single phase
AC input frequency	50–60 Hz
Maximum current	MAC 20 A DMC 10 A
Environmental	
Operating temperature	0–40°C
Storage temperature	-10–60°C
Operating humidity	5–95% RH non-condensing
Operating altitude	Below 2000 m (6562 ft.)
Mounting location	Inside a protected electronics rack only.
Remote communications	
Remote communication type	Ethernet or serial or analog (0–10V, Type AVP only)
Communications protocol	Modbus TCP, Modbus RTU (RS232) or Bisync (RS422, with Veeco Protocol converter)
IP Address	See table in the Appendix.
Software (GUI)	

Item	Specification
MAC2015	PC Win7 or Later Required for all applications. (Legacy tools using MAC/DMC require Molly upgrade, or Veeco Tablet Interface option, or a separate user-provided PC)
Tablet Interface (optional equipment)	Contact Veeco for details.
Software (process control)	
Veeco Molly® Growth Control Software	Required for all applications except automated valve positioning on a non-Veeco MBE system. See Molly version requirements.
Non-Veeco process control software	Contact Veeco for requirements.

Chapter 2 Introduction

Product Overview

About the MAC and DMC Controllers

The MAC and DMC Controllers are high-performance servo-motor controllers for Veeco Molecular Beam Epitaxy (MBE) equipment. The MAC controller can control up to eight axes of motion, and the DMC Controller can control up to two axes of motion (see [Applications](#) on page 9).

Features of the MAC and DMC Controllers include:

- Nested-loop servo motor control
- Motor control resolution of 4096 counts/revolution
- Out-of-the box support for eight types of motion, supporting future re-purposing
- PC-based, user-friendly graphical user interface (GUI)
- Seamless integration with the Veeco Molly Process Control Software and Lot Manager scheduling software
- Legacy support for Modbus RTU over RS232, Bisync on RS422 and Analog remote control *

The MAC and DMC Controllers are typically packaged with servo motor(s), cabling and mounting hardware.

* Analog remote control is available only for automated valve positioning. On the DMC Controller, both axes may be operated in analog mode; on the MAC Controller, up to four axes may be operated in analog mode.

Product Versions

The two versions of this product include the MAC Controller and DMC Controller, and differ as follows:

- MAC Controller:
 - 5U full-rack chassis
 - Configured with up to 8 axes of motion
- DMC Controller
 - 2U full-rack mount

- Configured with 1 or 2 axes of motion

Both product versions support Modbus TCP, Modbus RTU, Bisync, and analog remote control options. Additionally, both versions have the same performance characteristics, use the same user interface and screens and enable eight possible applications (see [Applications](#) on page 10).

Applications

The MAC and DMC Controllers can be used for the following applications. For details on each application, refer to the sections that follow.

- Automated valve positioning on Veeco Valved Crackers and other valved sources;
- Growth- or prep-stage rotation in a fixed, horizontal plane on Veeco systems with a vertically mounted substrate manipulator;
- Growth stage rotation in a variable plane on Veeco systems with a two-axis substrate manipulator;
- Growth stage/BFM indexing on Veeco systems with a two-axis substrate manipulator;
- Linear elevator (up/down) positioning in load-lock or storage-modules on Veeco systems;
- Linear positioning of a quartz crystal monitor assembly or a linear BFM;
- Robot arm motion, including rotation and extension, on automated Veeco systems; and
- Wafer flipping motion, including clamping and rotation, in a Veeco Wafer Flipper Module.

Identifying the Parts

Dual Axis Motion Controller (DMC Controller)

The DMC Controller is offered in a 2U full-rack form. It can be configured for 1 or 2 axes of motion.



Dual Axis Motion Controller (DMC Controller)

Multi Axis Motion Controller (MAC Controller)

The MAC Controller is offered in a 5U full-rack form. It can be configured for up to 8 axes of motion.



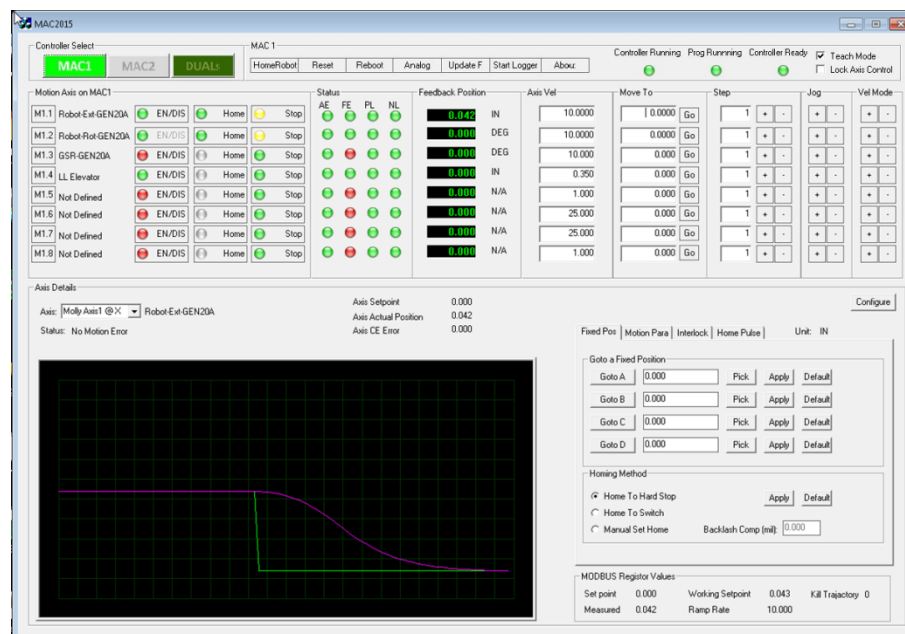
Multi Axis Motion Controller (MAC Controller)

GUI Software

MAC2015 is the GUI software for the MAC and DMC controllers. It is used to manage the controllers, configure new axes of motion, initiate axis motion and manage axis parameters. It also provides a more comprehensive way to control each axis of motion than what is typically available through remote process control software. The MAC2015 software can be started through use of a single

executable file, MAC2015.exe. By default, when the software is installed for a Veeco system, a shortcut to this file is placed on the desktop of the Molly computer.

The MAC2015 software is designed for Windows 7 or later operating systems.



MAC2015 Software

User Interface Options

The MAC2015 software can be loaded on any computer that is on the same network as the MAC or DMC Controller and is running Windows 7 or higher.

An optional tablet can also be used to run the MAC2015 software. The tablet can be mounted on the MBE system as shown below.



Optional Tablet

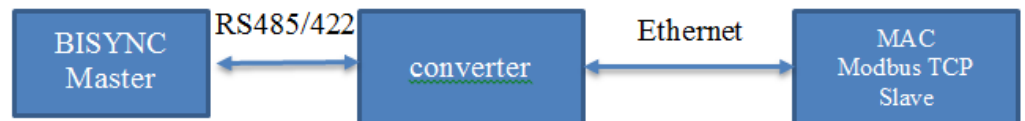
Protocol Converter Option

An optional protocol converter interface enables the use of the MAC and DMC Controller on Bisync systems.



Optional Protocol Converter Interface

The converter converts a MODBUS TCP slave device into a Bisync slave device.



TCP to Bisync Conversion Process

Chapter 3 Installation

Installation Overview

No installation is required for MAC or DMC Controllers purchased as part of a new Veeco MBE system. All procedures covered in this chapter are completed by Veeco either in the factory or as part of MBE system installation. However, you may need certain information in this chapter if you ever need to re-install the controller or re-commission it for a different application.

In most cases, installation is required only when the MAC or DMC Controller is purchased as an add-on or replacement unit for existing Veeco MBE equipment, or as part of a Veeco Valved Cracker package.

The overall installation process can be divided into four areas:

- Mechanical installation;
- Electrical integration;
- MAC2015 software installation; and
- Process control software configuration.

Refer to the sections that follow for installation instructions in each of these areas.



WARNING — HIGH VOLTAGE. Installation should be performed only by a trained MBE technician. Make sure that all sources of input power to this product are turned off and disconnected before installation. Do not attempt to service the inside of the controller. Beware of high voltage inside the electronics rack.

Mechanical Installation

Overview – Mechanical Installation

Mechanical installation of the MAC/DMC Controller involves:

- Installation of the controller in the electronics rack; and
- Installation of the servo motor assembly.

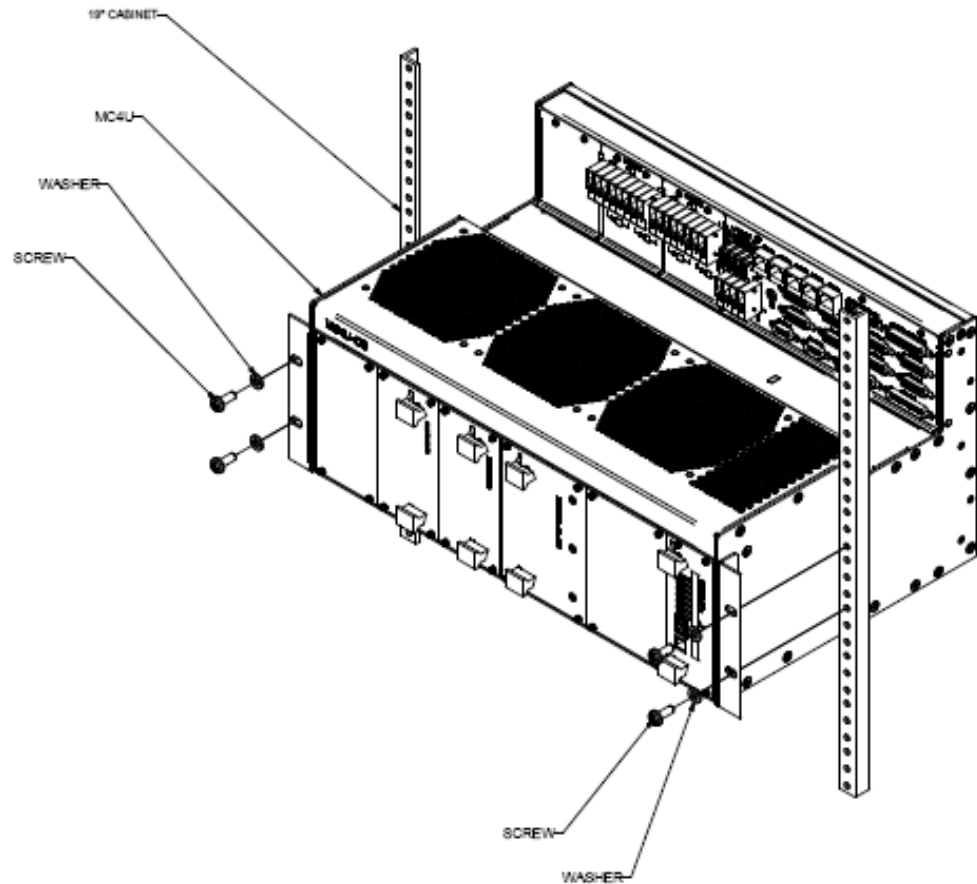
Refer to the sections that follow when performing these tasks.

Installing the Controller

The MAC Controller is designed to occupy an industry standard full-rack 5U (8.75" x 19") slot in a 19.0" equipment rack.

The DMC Controller is designed to occupy an industry standard full-rack 2U (3.5" x 19") slot in a 19.0" equipment rack.

If necessary, you can obtain exact mounting-hole dimensions from the equipment rack or the vendor of the rack.



Screw Locations for Rack Mounting

Installing the Servo Motor Assembly

The procedure for installing the servo motor assembly depends on the type of motion that the MAC or DMC Controller will be controlling.

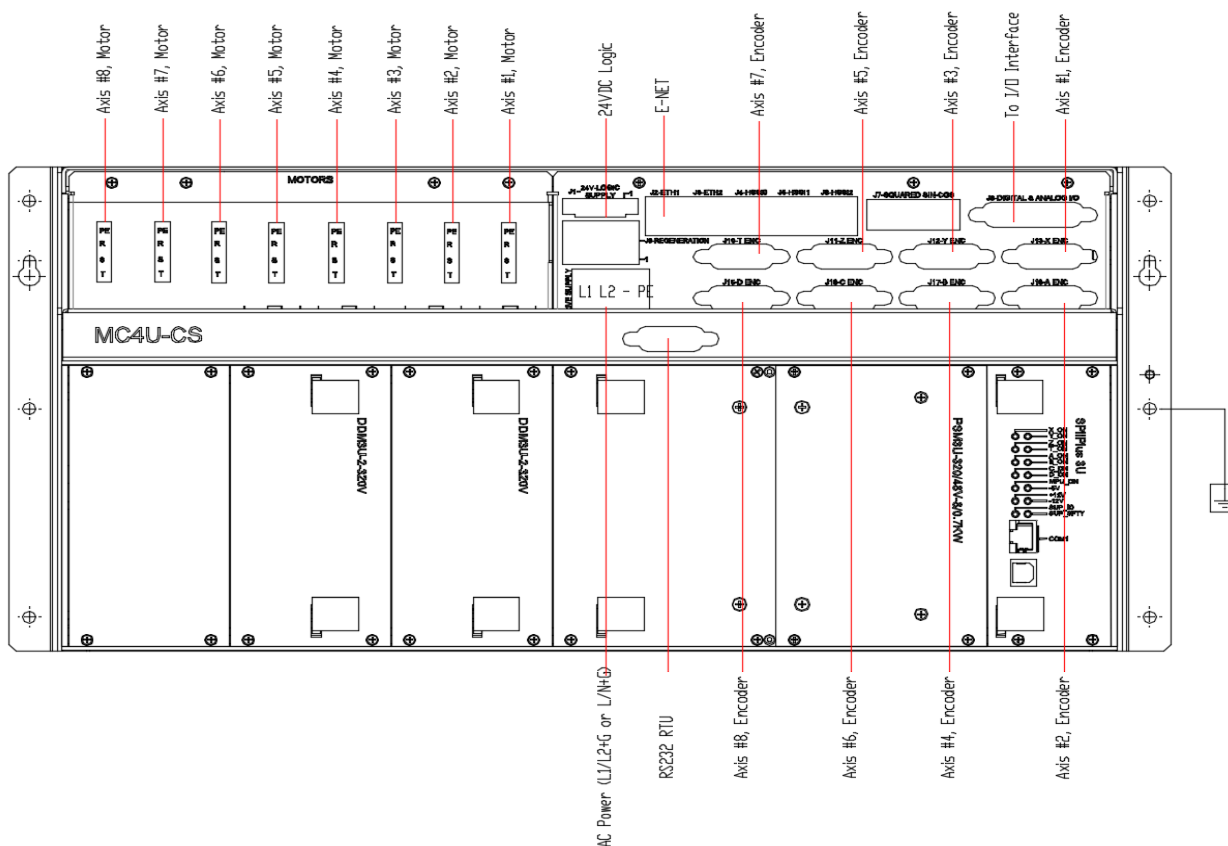
Several options exist for motor types, motor couplers, gear reducers, and motor mounts. If you need instructions for installing the servo motor assembly, please contact Veeco Technical Support.



CAUTION — Damage could occur if the wrong servo motor is accidentally installed onto a component. The hazard is greatest for axes of motion configured for automated valve positioning. To avoid damage, ensure that the original servo motor gets re-installed onto each component anytime multiple servo motors have been removed for system bake or maintenance.

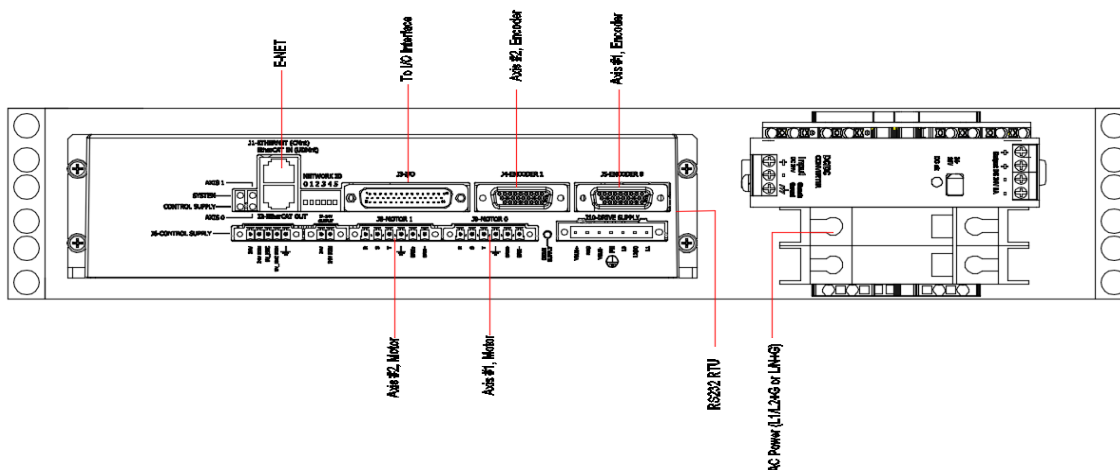
Electrical Integration

Electrical Integration - MAC Controller



Electrical Integration for the MAC Controller

Electrical Integration - DMC Controller



Electrical Integration for the DMC Controller

MAC2015 Software Installation

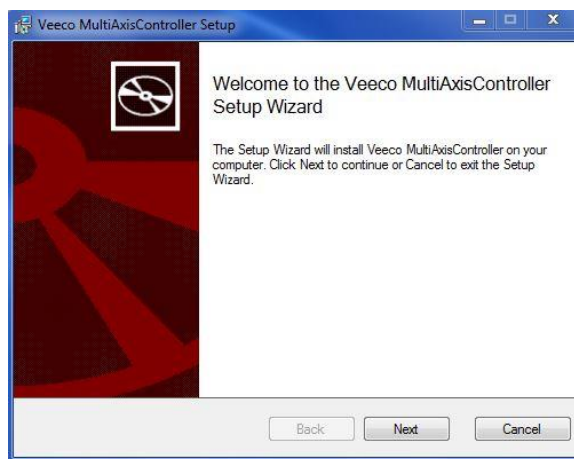
If you purchased the MAC or DMC Controller as part of a new Veeco MBE system, the MAC2015 software—which serves as the user interface for the controller—is pre-installed on the Molly computer. No further software installation is required.

If you purchased the MAC or DMC Controller as part of an upgrade or replacement for an existing Veeco system, or as part of a Veeco Valved Cracker package, you will need to install the MAC2015 software either on the Molly computer (in the case of a Veeco system upgrade), or on a compatible PC on the same network as the MAC or DMC controller (in the case of a non-Veeco system).

Use the following procedure to install the MAC2015 software.

1. On the USB flash drive provided with the controller, locate and open the following file: **1265589_IMAC User Interface installer.zip**
The VeecoMultiAxisController.exe file is displayed.

2. Run the **VeecoMultiAxisController.exe** file. The Veeco MultiAxisController Setup Wizard starts.



Installing the MAC2015 Software

3. Navigate through the Setup Wizard to install the required file(s).

After installation is finished, the MAC2015 software is located at **C:\Veeco\IMAC**.

Process Control Software Configuration

Overview - Process Control Software Configuration

If you purchased a MAC or DMC Controller as part of a new Veeco MBE system, the Molly Growth Control software comes pre-configured to communicate remotely with the controller. No further software configuration is required.

If you purchased a MAC or DMC Controller as an add-on or replacement unit for existing Veeco MBE equipment, or as part of a Veeco Valved Cracker package, you will need to configure your process control software to communicate with the controller. The procedure for doing this depends on whether it is a Veeco MBE system (and therefore uses some version of the Molly software) or a non-Veeco system (and therefore uses other process control software).

Refer to the applicable software section of this manual.

Software Configuration – Veeco Systems

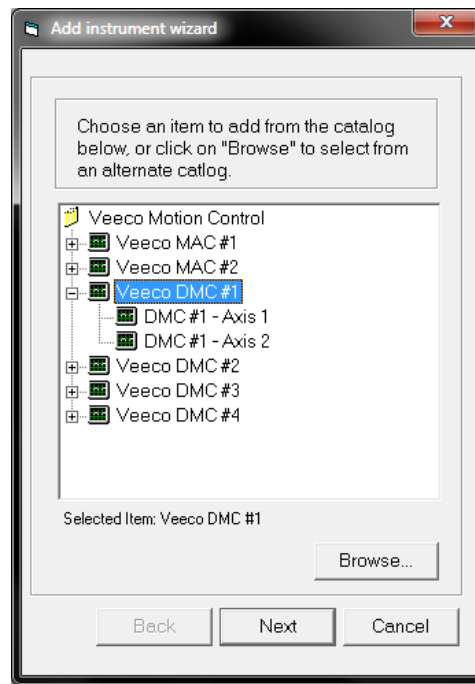
The Molly software must be configured to enable remote communication with the MAC or DMC Controller. Follow the steps below to configure Molly for the MAC or DMC Controller.

1. Insert the USB drive provided with the MAC or DMC Controller into a USB port on the Molly computer.

2. Open the text file 'InstallingPackages.txt' in the Veeco MAC Software folder on the USB drive and follow the installation instructions.



IMPORTANT — When selecting an instrument, be sure to select the correct instrument based on the growth module number, port number, module, etc. Once added, ensure that the system device IP address matches the address for the controller being installed.



Instrument Wizard Example Instruments

3. Add a zone(s) to the Molly configuration that references the instrument you just added.

TIP — Refer to Molly Software help for instructions on adding zone objects.

4. Save the configuration and restart the Molly user interface and kernel.
5. In Molly, test each axis of motion to ensure that remote operation of the controller is working properly.

Software Configuration – Non-Veeco Systems

NOTE — This section applies to a MAC or DMC Controller being used as an Automated Valve Positioner on a non-Veeco MBE system.

To configure non-Veeco process control software for remote communication with a MAC or DMC Controller being used as an automated valve positioner, you will need to select the necessary information from [Appendix A: Remote Connections Interface](#) on page 44, and build this into your software user interface and/or software kernel.

Chapter 4 Operation

Quick-Start the Controller

1. Make sure that the MAC or DMC Controller and servo motor assembly are properly installed.
2. Turn on the circuit breakers for the MAC or DMC Controller.

TIP — The MAC Controller is under two circuit breakers; one is for the 24V logic power supply, and the other is for motor power. See your system interconnection diagram to find the circuit breaker locations. The DMC Controller has a single, on-board circuit breaker on the back of the assembly. Just turn the circuit breaker on to power on the controller.

3. Wait approximately 20–30 seconds.
4. Start the MAC2015 software by double-clicking the MAC2015.exe file.

TIP — The MAC2015.exe file is located at C:\Veeco\MAC and may have a shortcut on the desktop.

5. Perform homing from the MAC2015 software.
6. After the homing is complete, the controller is ready to take motion commands from the MAC2015 software, or from properly configured remote process control software.

Motion Safety

Overview

The MAC and DMC Controllers must be incorporated into the emergency stop circuitry for the system. Refer to [Emergency Stop](#) on page 21 for details.

Depending on the MBE system configuration, PLC interlocks may be included to prevent mishaps during motion control that could potentially result in equipment damage. However, this is not always the case and may vary by system.

Before using the MAC or DMC Controller for actual positioning of components, review the [Hazards Table](#) on page 22 to learn about potential hazards and how to avoid them.



CAUTION — Failure to take the appropriate hazard-avoidance measures could result in equipment damage.

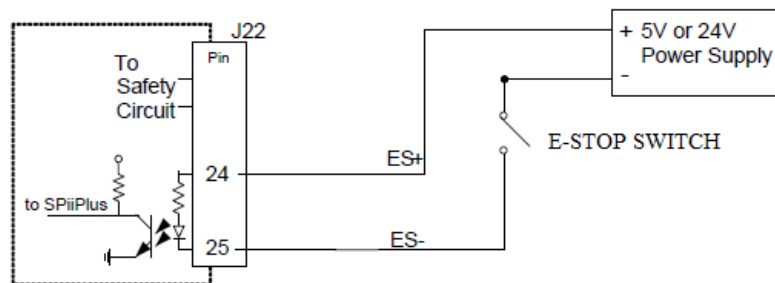
Emergency Stop

On Veeco MBE systems that include a MAC or DMC Controller, the emergency stop for motion is governed by the system EMO button.

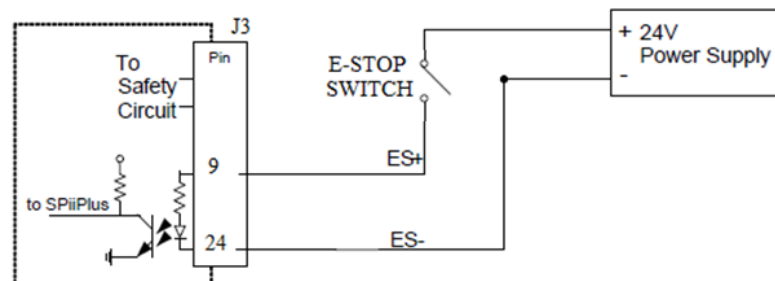
On non-Veeco systems, for motion safety, the customer is responsible for ensuring that power to the MAC and/or DMC Controller can be removed by the system EMO circuitry. If a dedicated motion emergency stop is required, both the MAC and DMC Controllers provide an interface for emergency stop input—refer to the connection diagrams below.



WARNING — EMERGENCY STOP DEVICE(S) REQUIRED. An emergency stop device must be located at each operator control station, and potentially other locations. The emergency stop device must disconnect any electrical equipment on the EMO circuit from the power supply, and require a manual reset by the operator. In situations with multiple emergency stop devices, the circuit shall not be restored until all emergency stop devices have been manually reset. Failure to include compliant emergency stop device(s) could result in serious injury or death.



MAC Controller Emergency Stop Input Interface



DAC Emergency Stop Input Interface

Hazards Table

The table below identifies hazards based on the application and lists recommended avoidance measures. (For an overview of the different [Applications](#), see page 10.)

Application	Hazard	Avoidance Measures
Automated valve positioning	The valve is accidentally left closed while the bulk-zone temperature is changed, resulting in needle-valve compression or damage. The valve is closed too forcefully on a Veeco Corrosive Series Valved Cracker. Controller is set to the wrong sub-type leading to incorrect direction, speed or torque limits. The valve is set at the wrong position for the process being performed.	Follow all guidelines and warnings about valve positioning in the applicable source, component, or system manual.
Growth- or prep-stage rotation in a fixed horizontal plane	The stage is rotated while the transfer arm is extended to the transfer position.	Always retract the transfer arm before initiating stage rotation. <i>NOTE—Veeco automated systems have built-in interlock features to prevent interference.</i>
Growth stage rotation in a variable plane.		
Growth stage/BFM indexing	The growth-stage/BFM assembly is indexed while other components (such as a transfer rod or QCM sensor) are extended into the path of travel.	Always retract any component whose path of travel can interfere with the growth-stage/BFM assembly before changing the target setpoint. Visually confirm that motion will be unobstructed before changing the target setpoint.
Linear elevator (up/down) positioning in load-lock or storage-modules	The module elevator is moved up or down too far during platen pick-ups or drop-offs, resulting in contact between the end effector on the transfer arm and other platens in the platen cassette.	Visually monitor all platen pick-ups and drop-offs. Do not use Fixed Positions for platen pick-ups and drop-offs
Linear positioning of a quartz crystal monitor assembly or a linear BFM	The sensor is extended into a closed isolation-chamber/growth-module gate valve. The sensor is extended into the growth stage or growth-stage/BFM assembly. The isolation-chamber/growth module gate valve is accidentally closed while the sensor is extended into the growth module.	Visually confirm that the gate valve is open before extending the sensor assembly into the growth module. Visually confirm that nothing is obstructing the sensor assembly's path of travel near the sensing position. Visually confirm that the sensor assembly is retracted past the gate valve before closing the gate valve.

Using the MAC2015 Software

Introduction

The MAC2015 software provides the graphical user interface (GUI) for MAC and DMC Controllers. It can simultaneously manage, and provide a single interface, for up to two MAC Controllers and up to four DMC controllers—a total of 24 axes of motion.

Password-Protected Features

Certain features in the MAC2015 software are password protected. To gain access, the following password must be entered: **4875**.

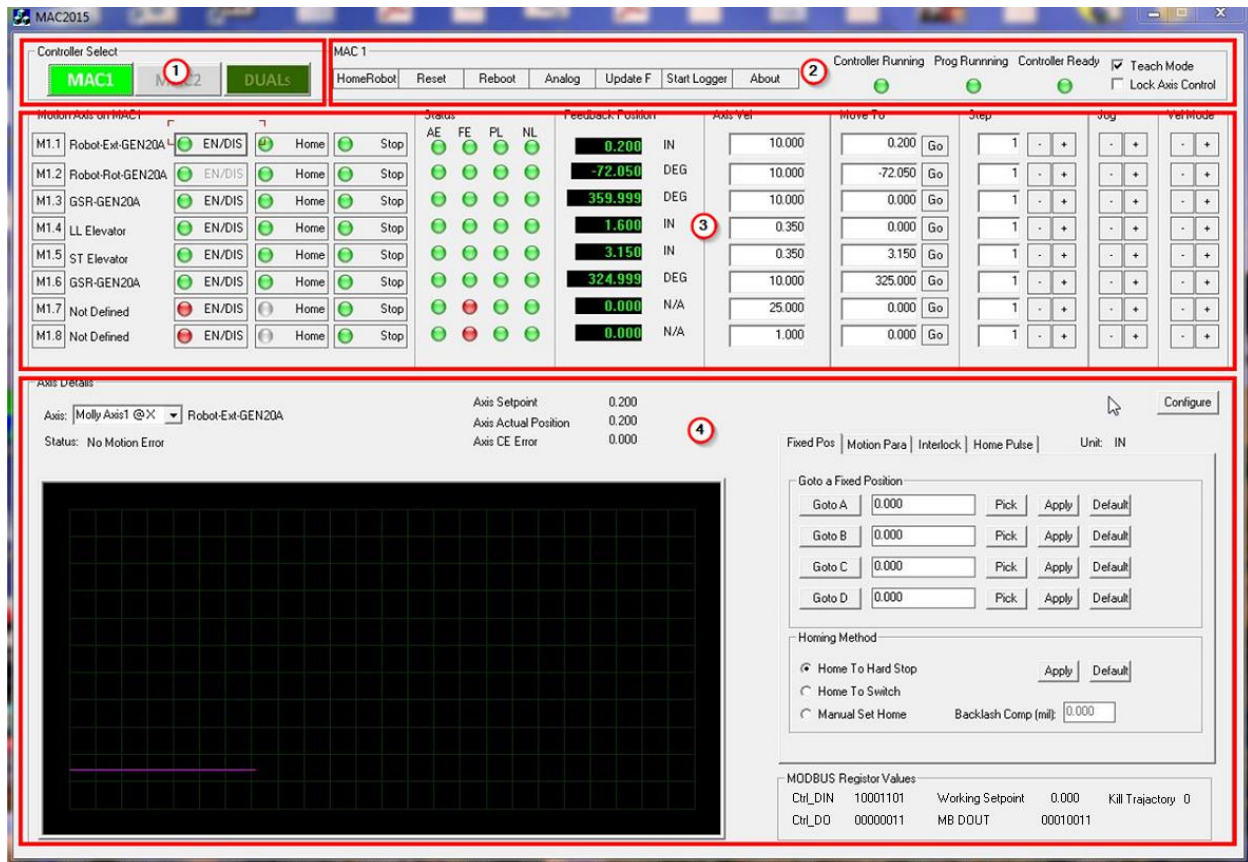


CAUTION — Changes to default values or settings on password-protected screens could result in equipment damage or poor performance. Do not change default values or settings without being certain that no harm can result.

Overview of the GUI

As shown in the figure on page 24, the MAC2015 software GUI is divided into four panels:

- Panel 1: **Controller Selection**
- Panel 2: **Controller Management** for the controller selected in Panel 1
- Panel 3: **Axis Control** for the controller selected in Panel 1
- Panel 4: **Axis Details** and configurations for the selected axis



MAC2015 GUI Window

Controller Selection Panel

In the Controller Selection panel (Panel 1), there are three Controller Select buttons: MAC1, MAC2 and DUALs. When the MAC2015 software starts up, it searches for all MAC and DMC controllers installed on the same network. The colors of the three buttons reflect the search result.



Controller Selection Panel

The meaning of different colors is as follows:

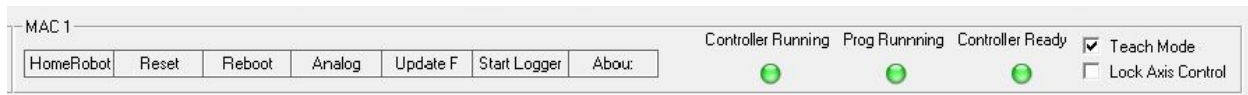
- **Light Green:** Current active controller. An active controller means that all the controls and information displayed in the GUI are for that controller. Example: MAC1 in the figure above.
- **Dark Green:** Installed controller(s), but currently not under active control. Example: DUALs in the figure above.

- **Grey:** Not installed or found.
Example: MAC2 in the figure above.



IMPORTANT — If multiple DMC Controllers are installed on the system, the controls and indicators in the MAC2015 software apply to a specific DMC Controller only after you both select the DUALs Controller Select button AND select an axis for that DMC Controller in the [Axis Control Panel](#) (see page 29).

Controller Management Panel



Controller Management Panel

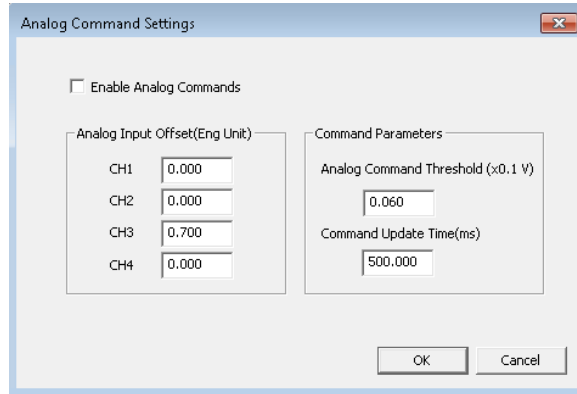
The Controller Management panel provides controller-level controls and information for the active controller (the controller selected in Panel 1). The name of the active controller is displayed in the top left corner of the panel.

The different features of the Controller Management panel are explained below.

- **HomeRobot** button. Performs robot auto-homing. This button is enabled only if two axes of motion are configured for a robot on that controller.
- **HomeFlipper** button (not shown in screenshot above). Performs flipper auto-homing. This button is enabled only for DMC Controllers dedicated to the two axes of motion for the flipper (flipper rotation and flipper clamping).
- **Reset** button. Restarts the controller firmware. Click this button after a parameter has been changed for an axis of motion; doing so reloads and initializes axis parameters. No re-homing of the affected axis of motion is required after a reset.
- **Reboot** button. Reboots the controller. After a reboot, you must re-home all axes of motion on this controller.
- **About** button. Displays information on the MAC2015 GUI version and the controller firmware version. If the MAC2015 GUI version is v2.08 or later, the Manufacturer's firmware version is also displayed.
- **Analog** button. Displays the Analog Command Settings window (see figure on page 26). Note the following:
 - The **Enable Analog Commands** box enables/disables analog remote control mode on the controller.
 - The **Analog Input Offset** is used to calibrate any zero offset. The value is in the Engineering Units that apply to that axis of

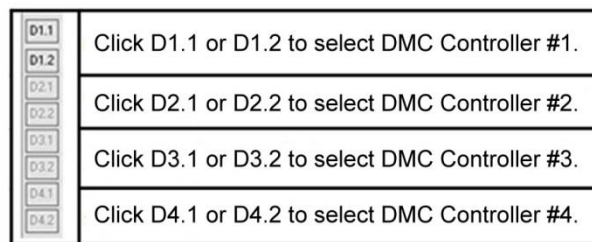
motion. It should be the number of the Feedback Position when the analog input is set to zero.

- The **Analog Command Threshold** value controls the sensitivity of the analog input. The unit is 0.1V.
- The **Command Update Time** value controls the sampling time interval of the analog command. The unit is ms.



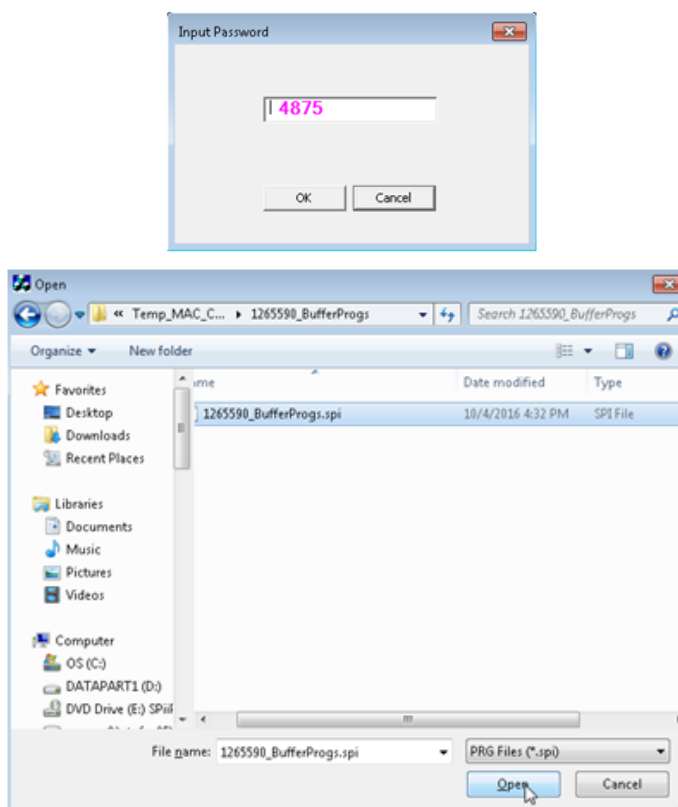
Analog Command Settings

- **Update Firmware** button. Updates the firmware (Veeco P/N 1265590) on the currently active MAC or DMC Controller. If multiple DMC Controllers are installed on the system, be sure to select the correct DMC Controller before downloading the firmware. To select the correct DMC Controller, click an axis of motion for that DMC Controller, as shown in the figure below.



Selecting a DMC Controller for Firmware Update

The Update Firmware function is password protected. After you enter the password, an Open file dialog is displayed, allowing you to open the new firmware file. Once the download is finished, you need to reboot the controller. Note that whenever you perform a firmware update, a firmware backup file is automatically saved to the same folder. This file can be loaded for recovery purposes in the event that a problem occurs during the controller firmware update. Note that you can also manually back up the firmware by clicking the main MAC2015 icon at upper left on the screen and choosing **Backup Firmware**.



Update Firmware Screens



IMPORTANT — If you have multiple MAC or DMC Controllers installed on your system, make sure that the correct controller is selected in the Controller Select Panel when updating the firmware. Also, when a firmware update is issued by Veeco, be sure to perform a separate firmware update on each MAC or DMC Controller installed on your system.



CAUTION — DISABLED AXES OF MOTION. Updating the firmware on a MAC or DMC Controller requires a controller reboot. All axes of motion will be disabled and require re-homing. Do not perform this operation during any system-critical processes.

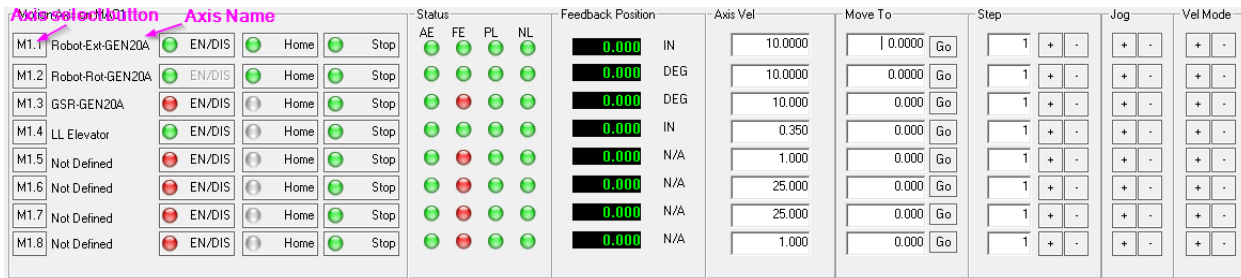
- **Start Logger** button. Enables the data-logging feature. The servo time stamp, setpoint and feedback position are recorded for each axis of motion on all the active controllers. The software generates a comma-separated-value file (.csv) for approximately every 10K data points for each active controller. The time interval between two samples is about 0.3 seconds. When this button is clicked, a 'Select Folder' dialog is displayed, letting you choose where the files are saved. Clicking the Start Logger button a second time turns off data logging.

- **Controller Running** indicator. Indicates the communication status of the currently active controller. Green indicates normal communication. Red indicates a communication problem.
- **Prog Running** indicator. Indicates the firmware running status of the currently active controller. Green indicates that the firmware is running normally. Red indicates a firmware problem. Grey indicates that the firmware running status cannot be determined.
- **Controller Ready** indicator. Indicates whether the currently active controller is ready to receive motion commands. Green indicates that the controller can take motion commands. Red indicates that motion commands cannot be taken. Grey indicates that it cannot be determined if motion commands can be taken.
- **Teach Mode** checkbox. Allows you to enter Teach Mode. Check this box before teaching the robot and uncheck it after teaching is complete. This checkbox is valid only if axes of motion are being used for robot control on the currently active controller.
- **Lock Axis Control** checkbox. Allows you to disable all control buttons for all axes of motion on the currently active controller. A password is required to re-enable the control buttons.



CAUTION — Make sure that the Teach Mode checkbox is unchecked after the robot is taught to enable robot arm rotation protection.

Axis Control Panel

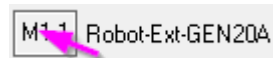


Axis Control Panel

The Axis Control panel provides functions for a single MAC Controller (controlling up to eight axes of motion) or for up to four DMC Controllers (each controlling up to two axes of motion).

The features and functions of the Axis Control panel are as follows:

- Axis Select** button. Shows the axis number (for example, M1.1 means Axis #1 on MAC#1, D2.1 means Axis#1 on DMC#2, etc.). Click an Axis Select button to display details for that axis in the Axis Details panel (Panel 4) at the bottom of the screen.



Axis Select Button

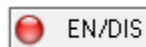
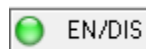
- Axis Name** field. Displays the name of the axis and allows you to customize the name. When you double-click the axis name, an edit box appears, allowing you to change the name. To exit editing mode, click the Axis Select Button. New axis names are saved automatically. This feature can be useful for differentiating two or more axes of motion with the same configuration. For example, if two axes have been assigned the S-500 configuration for automated valve positioning, you could change the axis names to S-500-Port#A and S-500-Port#B.



Axis Name

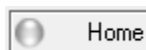
- **EN/DIS** button. Allows you to enable or disable an axis of motion by clicking the button. Also has an indicator light that shows the enabled/disabled motor feedback status: green means the motor is enabled, and red means the motor is disabled.

NOTE—On automated systems, the enabled/disabled status for the two robot axes of motion are controlled by a single button.



Enable/Disable Button

- **Home** button. Allows you to home the axis of motion using the homing method (manual homing, homing to hard stop, or homing to switch) currently defined for that axis—see [Axis Details panel](#) on page 32. Also has an indicator light that shows the homed/not-homed feedback status: green means the axis has been homed, and grey means the axis has not been homed.



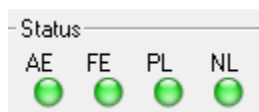
Home Button

- **Stop** button. Allows you to stop motion for an axis currently in motion. Also has an indicator light that shows the motion status for the axis: yellow means the axis is in motion, and green means the axis is stopped at a position.



Stop Button

- **Status** indicators. Provide feedback status for potential axis errors. Green indicates a good status (no error). Red indicates a bad status (active error).



Status Indicators

- **AE** – Amplifier Error.

- **FE** – Feedback Error. Usually this error indicates that the encoder cable is disconnected.
- **PL** – Positive limit status. If the positive limit is reached, the controller will only allow motion away from the positive limit.
- **NL** – Negative limit status. If the negative limit is reached, the controller will only allow motion away from negative limit.
- **Go** button. Starts an axis movement toward a target setpoint (the Move To value) at the specified velocity (the Axis Vel value). If the axis is idle when the command is issued, the motion starts immediately. If the axis is in motion when the command is issued, the motion will transition to this command immediately and end up at the Move To position.

The screenshot shows a control interface with two input fields. The first field is labeled 'Axis Vel' and contains the value '20.000'. The second field is labeled 'Move To' and contains the value '0.010'. To the right of the 'Move To' field is a button labeled 'Go'.

Go Button, Axis Vel Field, and Move To Field



IMPORTANT — If the axis is being used for growth- or prep-stage rotation, the velocity units are RPM.

- **Step** buttons (+/-). These issue a step command with the specified step size in the positive (+) or negative (-) direction. When the command is issued, the motion starts immediately. Each click results in one step.

The screenshot shows a control interface with a label 'Step' above a numeric input field containing the value '1'. To the right of the input field are two buttons labeled '-' and '+'.

Step Buttons

- **Jog** buttons (+/-). When clicked and held, these buttons start a motion in the positive (+) or negative (-) direction with constant velocity (defined by Axis Vel) and no defined endpoint. The motion continues until the mouse button is released, a limit switch is activated, or another condition preventing motion occurs.

The screenshot shows a control interface with a label 'Jog' above two buttons labeled '-' and '+'.

Jog Button

- **Vel Mode** buttons. When clicked once, these buttons start a continuous motion in the positive (+) or negative (-) direction with constant velocity (defined by Axis Vel) and no defined endpoint. The motion starts immediately when the button is clicked and continues until the Stop button for that axis is clicked, a limit switch is activated, or another condition occurs that prevents motion.

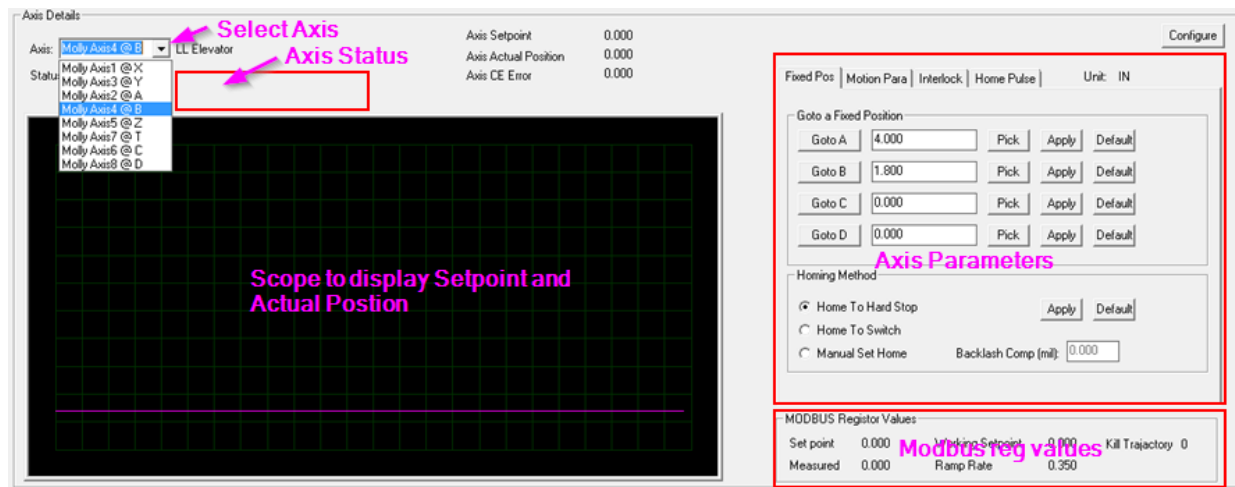


Velocity Mode Command Button

Axis Details Panel

Overview

The Axis Details panel provides detailed status, configuration and control options for the axis of motion selected on the active controller.



Axis Details Panel

Fixed Positions Tab

The Fixed Positions tab lets you command the component to move to any of four fixed positions. These are pre-defined setpoints for common functional positions.

Fixed Positions Tab

By default, there are four pre-defined fixed positions for each type of motion. The positions are displayed in the edit boxes.

- **Goto X** button. Issues a command to move the component to the defined position.
- **Pick** button. Sets the current actual position of the component as the new fixed position.
- **Default** button. Recalls the default value for the fixed position.
- **Apply** button. Saves any changes made.



CAUTION — See [Motion Safety](#) on page 20 for a review of possible obstructions to watch out for when moving a component.

Motion Parameters Tab

The Motion Parameters tab allows you to modify the default values for the selected motion axis. In some cases, this may be necessary to optimize performance. However, it is not normally recommended or required. Please contact Veeco Technical Support if you have questions about modifying the default values.



CAUTION — Changes to default values for motion parameters could result in equipment damage or poor performance. Do not change default values without being certain that the resulting control behavior cannot harm your equipment.

The two buttons on the Motion Parameters tab function as follows:

- **Default** button. Recalls the default value defined for the parameter.

- **Apply** button. Saves any changes made. Note that any change made is not effective until the controller has been reset.

The screenshot shows the 'Motion Parameters' tab. At the top, there are tabs for 'Fixed Pos', 'Motion Para', 'Interlock', and 'Home Pulse', with 'Unit: IN' on the right. The 'Motion Parameters' section contains a list of parameters, each with a text input field, an 'Apply' button, and a 'Default' button.

Parameter	Value	Apply	Default
Positive Limit	10.000	Apply	Default
Negative Limit	-0.010	Apply	Default
Move Vel	0.350	Apply	Default
Jog Vel	0.200	Apply	Default
Vmax	5.000	Apply	Default
Amax	1.000	Apply	Default
Fault Torq (%)	72.000	Apply	Default
Home Torq (%)	10.000	Apply	Default
Home Vel	0.100	Apply	Default
Home Offset	0.000	Apply	Default

Motion Parameters Tab



IMPORTANT — If the axis is being used for growth- or prep-stage rotation, the velocity units are RPM.

Interlock Tab

The Interlock tab allows you to modify the default values that determine how an axis of motion behaves when an interlock and/or 2nd setpoint is activated.

The screenshot shows the 'Interlock' tab. At the top, there are tabs for 'Fixed Pos', 'Motion Para', 'Interlock', and 'Home Pulse', with 'Unit: IN' on the right. The 'Interlock' section contains several sub-sections with various settings.

Interlock Options

☐ None ☒ Intlk by DIN2 ☐ Intlk by GMIU Apply All

Interlock Behavior

☐ Go to Interlock Position ☒ Stop Motion Default

Interlock Position Setting

Intlock Position CAR Intlk Pos

2nd SP Settings

☐ Enable 2nd SP
 2nd SP Position

☐ Inverse Logic (Checked: 0=Interlocked; Unchecked: 1=Interlocked)

Interlock Tab

- **Interlock Options** group. Defines the signal that interlocks the component.
 - **None** option – No interlock for the component.

- **Intlk by DIN2** option – Interlock is triggered by the rising edge of a signal presented at DIN2 (digital input #2).
- **Intlk by GMIU** option – Interlock is triggered by a growth-module PLC signal that is active when the component's path of travel is obstructed by another growth-module component. This option is available only for an axis of motion configured for linear positioning of an in-situ monitoring device (quartz crystal monitor [QCM] or beam flux monitor [BFM]) or an ozone injector.
- **Interlock Behavior** group. Defines how the axis of motion behaves when the interlock is triggered.
 - **Go to Interlock Position** option – Component is moved to the setpoint entered in the Intlock Position field. This option is a factory preset and is mandatory if the axis of motion is configured for linear-positioning of an in-situ monitoring device (QCM or BFM) or an ozone injector.
 - **Stop Motion** option – Component is stopped in place if currently moving or prevented from moving if stationary.
- **Interlock Position Setting** group. Defines the position(s) to which the component is moved if certain conditions are met (see below).
 - **Intlock Position** field – Value defines the position to which the component is moved if the interlock is triggered and the Go to Interlock Position option is selected. This value is pre-set in the factory for an axis of motion configured for linear-positioning of an in-situ monitoring device (QCM or BFM) or an ozone injector. The value must fall within the range of motion for the axis and be safe given the state of the system after the interlock is triggered — for example, for a QCM, it must be safely outside the growth module because the QCM gate valve closes.
 - **CAR Itlk Pos** field – Value defines the position to which the component is retracted if currently obstructing the growth stage while the latter is commanded to the Down position. This field is only used (and includes a factory preset value) for an axis of motion configured for linear-positioning of an in-situ monitoring device (QCM or BFM) or an ozone injector.
- **2nd SP Settings** group – Defines how the axis of motion behaves when the 2nd setpoint signal is active.
 - **Enable 2nd SP** checkbox – Turns the 2nd setpoint behavior on or off.
 - **2nd SP Position** field – Value defines the position to which the component is moved when the rising edge of a signal presented at DIN4 (digital input #4).

- **Inverse Logic** checkbox – Reverses the logic for the DIN2 and DIN4 signals as they relate to triggering of interlocks.

Home Pulse Tab

The Home Pulse tab allows you to modify the default values that determine how an axis of motion behaves with regard to home pulse output for reflection high-energy electron diffraction (RHEED). The output is a 24V signal.



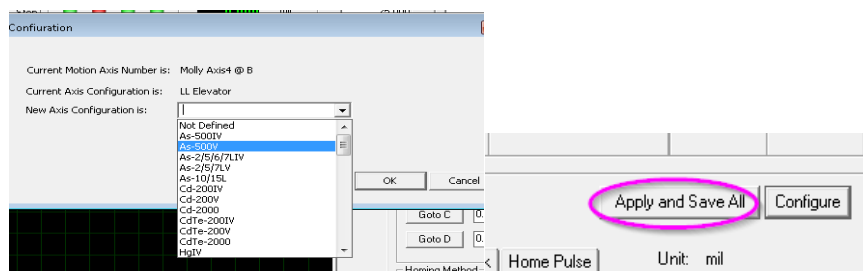
Home Pulse Output Tab

This tab has two parameters that define the pulse properties:

- **Width (% Turn)**. This value defines the width of the output pulse as a percentage of one revolution (360°) of the stage.
- **Delay (% Turn)**. This value defines the location of the starting edge of the pulse as a percentage of one revolution (360°) of the stage.

Configure Button

The Configure button allows you to change the configuration (application) for an axis of motion. After selecting a new configuration, click **Apply and Save All**. The new configuration takes effect after you reset the controller.



Axis Configuration



CAUTION — When the configuration is changed for an axis of motion, the MAC or DMC Controller must also be re-commissioned for that axis of motion. Contact Veeco Technical Support for details.

Homing an Axis of Motion

Homing Overview

Homing assigns a value of 0 (“home”) to a specific position in the range of travel for the component. The home position serves as a reference point that allows safe and repeatable control of a component’s motion.

You must home an axis if any of the following are true:

- the indicator light on the Home button for an axis appears grey (for any reason) in the Axis Control panel in the MAC2015 software;
- the MAC or DMC Controller has just been powered on for the first time;
- the MAC or DMC Controller has been power-cycled (ON > OFF > ON), either intentionally or due to a loss of power;
- the servo motor(s) or gear box(es) have been removed or decoupled from the system since the last homing (typically, removal/installation of these components occurs only for system bake or maintenance);
- the MAC or DMC Controller has experienced a fault; or
- (for automated valve positioning only) temperature changes have been made to one or more heating zones on the Veeco Valved Cracker or valved source (please consult the appropriate source user manual for details).



CAUTION — Potential equipment damage. See [Motion Safety](#) on page 20 for a review of possible obstructions to watch out for when moving components during homing.

Types of Homing

There are three types of homing: homing to hard stop, homing to switch and manual homing.

The correct homing method needs to be defined in the Axis Details panel (see figure below and refer to the sections that follow.)

Homing Method

Homing to Hard Stop

During homing to hard stop, the MAC or DMC Controller commands the servo motor to move the component in the negative direction until a physical limit is reached. As the limit is reached, the torque on the servo motor shaft suddenly increases. When the torque equals the pre-defined Homing Torque value, the MAC or DMC Controller assigns a value of 0 ("home") to the corresponding position.

Homing to hard stop is possible when the travel range has physical limits corresponding to a specific function; for example, the needle valve being fully closed on a Valved Cracker, or the elevator being all the way down in a load-lock module. For these components, homing to hard stop is recommended because it produces consistent home positions that are also safe due to the use of the pre-defined Homing Torque which prevents over-torque conditions.

Homing to hard stop is recommended when the MAC or DMC Controller is configured for automated valve positioning, linear elevator (up/down) positioning in load-lock and storage modules (except on GEN10 systems), robot extension, and flipper-module clamping.

Homing to Switch

During homing to switch, the MAC or DMC Controller commands the servo motor to move the component in the negative direction until a physical switch is reached. The switch is connected to the digital input of the MAC or DMC Controller.

Homing to switch is required for axes of motion that have been configured for robot rotation, linear elevator (up/down) positioning in a GEN10 load-lock

module, or linear positioning of a quartz crystal monitor assembly or a linear BFM.

Manual Homing

During manual homing, the operator—while carefully observing—manually moves the component to a specific functional position, and then presses a button on the MAC or DMC Controller to set this position as 0 (“home”).

Manual homing is required for components that either have no physical limits in the range of travel or have physical limits that do not correspond to a useful function. For example: in the case of the growth stage on a vertically mounted substrate manipulator, “home” should be the position at which the legs of the growth stage do not interfere with the platen-transfer axis; in the case of an indexed growth-stage/BFM assembly, “home” should be the position at which the growth stage precisely faces the sources.

Manual homing is required for axes of motion that have been configured for growth- or prep-stage rotation on non-automated Veeco systems, or for growth stage/BFM indexing on non-automated Veeco systems.

Homing in Automated MBE Systems

About Homing the Robot Axes

It is recommended that you use the auto-home sequence (HomeRobot button) when homing the robot. The auto-home routine begins by homing the robot extension axis to a hard stop. Next, it homes the robot rotation axis to the robot rotational home switch—a photo eye mounted to the top of the cluster module near the cluster/growth transfer port.



CAUTION — On automated systems, after robot auto-homing, be sure to visually confirm that the robot’s rotational photo-eye laser is centered on the protruding “flag” on GEN20A, GEN200 and GEN2000 systems, or on the “neck” of the end effector on GEN10 systems.

About Homing the Growth Stage

On automated systems, the growth- and prep-stage rotation axes are homed to switch. This can be done by clicking the **Home** button for the axis in the MAC2015 software, or by using the **Set Rotation Home** (auto-home) function in the Lot Manager software before teaching the growth- or prep-module transfer positions for the robot.



CAUTION — On all automated systems, after the growth- and prep-stage rotation axes are homed, always visually confirm that the stage is properly aligned at the transfer position to avoid a collision with the robot.

About Homing the Flipper (If Present)

The flipper module has two axes of motion: clamping and rotation. The motion of these two axes is always controlled and coordinated by a dedicated DMC Controller.

There are multiple ways to home the flipper. Two procedures are given below. These vary based on whether the flipper clamping axis is homed to hard stop or manually homed, and whether a platen is present or not.

Procedure 1 (Flipper Clamping Axis Homed to Hard Stop)

1. Make sure that the robot axes, load-lock elevator axis, and storage elevator axis (if present) are already homed.
2. Determine the correct Home Offset value, as follows:
 - If the flipper is holding a platen, use .03 inch.
 - If the flipper is empty (no platen), measure the thickness of the platen edge (see figure below), and then add .03 inch.



Platen Edge

3. In the MAC2015 software, on the Axis Details panel for the flipper clamping axis, enter the value from step 2 into the Home Offset field (on the Motion Parameters tab).
4. Click the **HomeFlipper** button. Both axes of flipper are homed: the flipper rotation axis is homed to switch, and the flipper clamping axis is homed to hard stop.
5. Verify the homing results.

Procedure 2 (Flipper Clamping Axis Manually Homed)

1. Make sure that:
 - the robot axes, load-lock elevator axis, and storage elevator axis (if present), are already homed; and
 - the robot is holding a platen.
2. Rotate the robot to the correct rotational angle for transferring to the flipper module, leaving the robot arms retracted.
3. Set the homing method for the flipper clamping axis to Manual Set Home and click Apply.
4. Make sure that the homing method for the flipper rotation axis is set to Home to Switch and click Apply.
5. Click the **HomeFlipper** button to perform homing of the flipper module.

When the homing process is finished, the home position for the flipper clamping axis is set to wherever the flipper clamping arms were, and the home position for the flipper rotation axis is at the rotation home sensor position. And you are now able to move the two axes of motion for the flipper.



CAUTION — The two axes of motion for the flipper motion are not yet at the true homes with respect to the robot end effector!

6. Using the step motion function in the MAC2015 software, drive the flipper clamping axis to fully open.
7. Slowly extend the robot arm into the flipper module while watching the motion through the viewports on the flipper module.
8. Slightly adjust the robot rotation and extension until the center of the platen is approximately aligned with the flipper clamp braces.
9. Again using the step motion function in the MAC 2015, slightly close the flipper clamping axis.
10. If necessary, as the flipper clamp braces approach the platen edge, slightly adjust the robot rotation and extension, as well as the flipper rotation, for proper centeredness and alignment.
11. When the robot rotation and extension are correct for transfer with respect to the flipper clamp braces, teach the robot for the flipper transfer position.
12. Continue to close the flipper clamping axis. The clamping arms will grip the platen, pick up the platen and finally be closed holding the platen.
13. Fully retract the robot arm.

14. If, in the above steps, you slightly adjusted (typically, by up to several degrees) the flipper rotation axis to achieve proper rotational alignment with the platen on the robot end effector, enter the Home Offset for the flipper rotation axis, as follows.

- a. Read the adjustment from the MAC2015 software. This is the Feedback Position value (in the Axis Control Panel) for the flipper rotation axis
- b. Enter this value in the Home Offset field for the flipper rotation axis. The Home Offset field is found in the Axis Details panel for the flipper rotation axis, on the Motion Parameters tab.
- c. Click Apply.

NOTE— The Home Offset value accounts for the difference between the flipper rotation at the home sensor position, and at the robot end effector.

15. Click the **HomeFlipper** button in the MAC2015 software to perform homing of the flipper module again.

16. Verify the homing results.

Chapter 5 Maintenance

Powering the Controller OFF

Powering OFF the controller is required only in preparation for factory service, or if the system is being taken out of service for an extended period.

Use this procedure to power OFF the MAC or DMC Controller.

1. Make sure that no application-critical processes (transfer, growth, monitoring) are occurring.
2. Switch off the circuit breakers from the electronics rack.

Powering the Controller ON

Use this procedure to power ON the MAC or DMC Controller.

1. Make sure that the MAC or DMC Controller, and the servo motor(s), are properly installed.
2. Switch on the circuit breakers from the electronics rack.
3. Wait 30 seconds.
4. Start the MAC2015 software.
5. Begin the homing process for all axes of motion.

Replacing Parts

The MAC and DMC Controllers contain no standard field-serviceable parts. Please contact Veeco Technical Support if you believe parts require replacement.

Removing the Controller

Removal of the controller may be necessary if factory maintenance is required.

Do not attempt to remove the controller from the electronics rack without contacting Veeco first. Veeco Technical Support will provide removal instructions if necessary.

Appendix A Remote Connections Reference

This Appendix provides information that is needed if you plan operate the MAC or DMC Controller remotely through non-Veeco process control software. In some cases, you may also need this information when configuring the Molly software for an existing MBE system with which the MAC or DMC Controller will be used.

Communications Protocol Overview

Controller IP Addresses

Default IP Addresses

The table below lists the default IP address settings for MAC and DMC Controllers in the same electronics rack.

MAC/DMC Controller	IP Address
1 st MAC Controller	192.168.1.210
2 nd MAC Controller	192.168.1.211
1 st DMC Controller	192.168.1.212
2 nd DMC Controller	192.168.1.213
3 rd DMC Controller	192.168.1.214
4 th DMC Controller	192.168.1.215

Changing a Controller IP Address

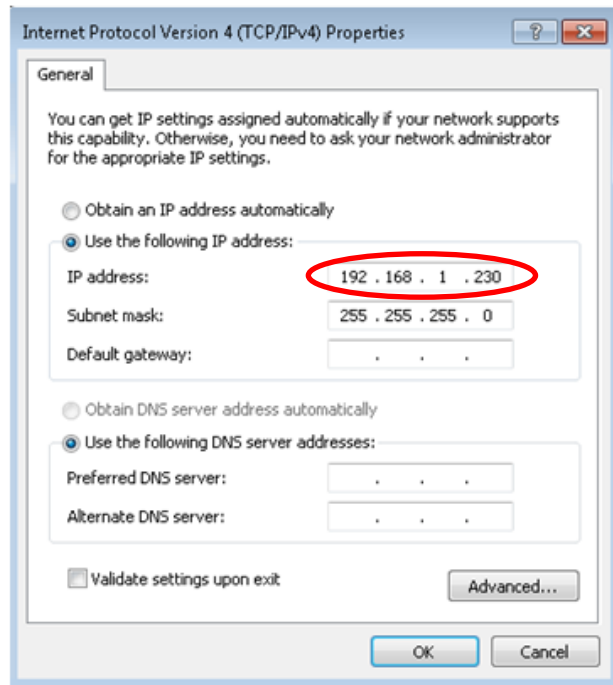
It is not necessary to change the IP address of a MAC or DMC controller installed as part of a complete Veeco MBE system.

If a MAC or DMC Controller is being used with a non-Veeco MBE system for automated valve positioning on one or more Veeco Valved Cracker(s), you may need to change the controller IP address first.

Use this procedure to change the controller IP address.

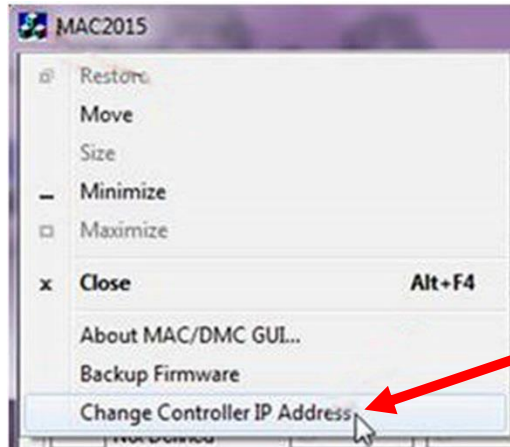
1. Prepare to update the controller IP address.
 - a. On the computer with the MAC2015 software installed, make sure that the Local Area Network (LAN) IP address for TCP/IPv4

is on the same network as the MAC/DMC Controller — for example, 192.168.1.230

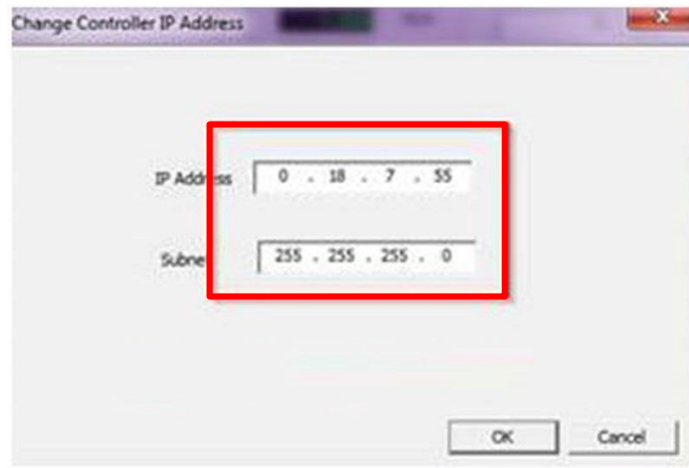


- b. Disconnect — at the *equipment network switch* — the Ethernet cable that runs from the computer with the MAC2015 software installed to the equipment network switch.
- c. Reconnect the Ethernet cable directly to the MAC/DMC Controller.

2. Update the MAC/DMC Controller IP address using the MAC2015 software.
 - a. Start the MAC2015 software. The MAC/DMC Controller is found if the proper Ethernet connection has been made.
 - b. Click the icon at upper left in the MAC2015 window and select Change Controller IP Address.

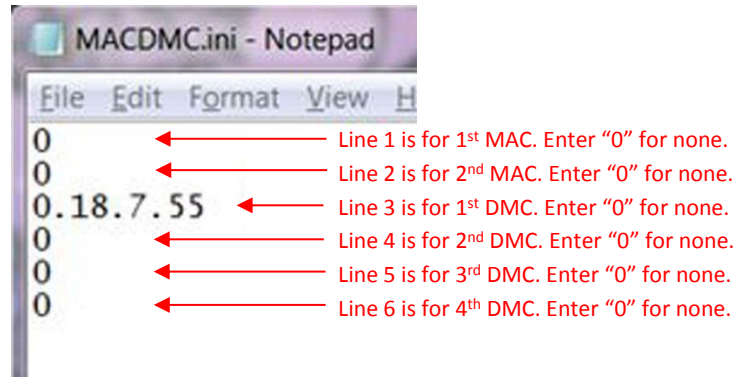


- c. Enter the new IP address for the controller — for example, 0.18.7.55



- d. Click OK.
 - e. Wait as the controller is updated, after which the MAC2015 software closes automatically.
3. Disconnect the Ethernet cable from the MAC/DMC Controller and reconnect it to the equipment network switch.
4. On the computer with the MAC2015 software, change the IP address for TCP/IP v4 to a new address on the same network as the MAC/DMC Controller's new address — for example, 0.18.7.230

5. Using a text editor, create a file like the one shown below and save it as **MACDMC.ini** in the same folder where the MAC2015.exe file is currently found. The new MAC/DMC Controller IP address should be included in the appropriate line, as shown in the example below.



6. Start the MAC2015 software again. The MAC/DMC Controller should now be found at its new IP address.

Modbus Tag Sets (Registers)

In the Modbus communications protocol, data are transferred in binary form and in arbitrary size (up to 254 bytes in a single transaction). Typically, tags are identified by "register numbers," where a register number denotes an offset into the data space of the device. A tag may be of "word" (16 bit) size, or of "float" (32 bit floating point) size. Multiple tags may be read at once, by requesting data to be read starting at the lowest register number, and by specifying a size large enough to encompass all the required data.

For the MAC and DMC Controllers, the serial port settings for Modbus are 8-N-1 (8 data bits, no parity, 1 stop bit).

Additional information on Modbus standards and specifications can be found at modbus.org, or from Veeco upon request.

The tables below list the base Modbus register addresses for each axis number on the MAC and DMC Controllers.

Modbus Base Registers - MAC Controller

NOTE — If the controller has a Manufacturer's firmware version before 2.6, remove the "4" at the beginning of the base Modbus register shown below: for example, 8400 instead of 48400.

Axis Number	Base Modbus Register
1	48000
2	48200
3	48400

4	48600
5	48800
6	49000
7	49200
8	49400

Modbus Base Registers - DMC Controller

NOTE — If the controller has a Manufacturer's firmware version before 2.60, remove the "4" at the beginning of the base Modbus register shown below: for example, 8400 instead of 48400.

Axis Number	Base Modbus Register
1	48000
2	48400

Bisync Tag Sets (Mnemonics)

In the Bisync communications protocol, data are transferred one tag item at a time. The data format is either integer or floating point. A tag is identified by a mnemonic, which is typically a two-character string. The first character of this string is almost always an alphabetic character; the second character may be alphabetic or numeric.

For MAC and DMC Controllers, the serial port settings for Bisync are 7-E-1 (7 data bits, even parity, 1 stop bit).

Additional information on Bisync standards and specifications are available at eurotherm.com or from Veeco upon request.

Data Tags for Modbus and Bisync

Supported Data Tags

The table that follows shows the data tags supported by MAC and DMC Controllers for the Modbus (RTU or TCP) and Bisync communication protocols. You may need this information if installing a MAC or DMC Controller on a system that does not use the Veeco Molly Growth Control Software.

For Modbus registers, please note the following:

- All registers are 32 bit
- Byte order = CD AB
- Bit format for DIN = BIN(CD) BIN(AB)



IMPORTANT — MAC and DMC Controllers that have been integrated with the Molly Growth Control software do not use all these data tags. Therefore, unused tags have not been tested and are provided for informational purposes only. Testing of all data tags is highly recommended before use.

Data Tag Name	Direction	Data Type	Modbus Register ⁽¹⁾	Bisync Mnemonic	Notes
Input Status	Read only	Integer	Base register ⁽²⁾ for axis + 0	–	–
Instrument ID	Read only	Integer	Base register ⁽²⁾ for axis + 2	–	–
Software Version	Read only	Integer	Base register ⁽²⁾ for axis + 4	–	–
DIN	Read only	Integer	Base register ⁽²⁾ for axis + 6	–	–
Output percentage	Read only	floating point	Base register ⁽²⁾ for axis + 8	OP ⁽³⁾	100 * Abs(CurrentPosition - MinimumTargetSetpoint) / Abs(MaximumTargetSetpoint - MinimumTargetSetpoint)
Stage home sensor at bit 0	Read only	Integer	Base register ⁽²⁾ for axis + 10	–	Applies only to growth- or prep-stage rotation.
Status Word	Read/write	Integer	Base register ⁽²⁾ for axis + 34	–	–
Kill Trajectory	Read/write	integer	Base register ⁽²⁾ for axis + 36	KT ⁽³⁾	If this tag is set to 1, the current executing movement is stopped in place. Tag automatically clears.
Measured Position	Read only	floating point	Base register ⁽²⁾ for axis + 100	PV ⁽³⁾	Feedback position. In Molly, this is the “Measured” value. Units = mil for automated valve positioning.

(1) All registers are 32 bit. Byte order = CD AB. Bit format for DIN = BIN(CD) BIN(AB).

(2) See [Modbus Base Register](#) tables on page 47.

(3) Bisync is an optional protocol for automated valve positioning only.

(4) It is recommended that changes to this parameter be made through the controller GUI.

Working Setpoint	Read only	floating point	Base register ⁽²⁾ for axis + 102	SP ⁽³⁾	Instantaneous target position. In Molly, this is the “Wsetp” value.
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Data Tag Name	Direction	Data Type	Modbus Register ⁽¹⁾	Bisync Mnemonic	Notes
Target Setpoint	Read/write	floating point	Base register ⁽²⁾ for axis + 134	SL ⁽³⁾	Position command for current move.
Target Velocity	Read/write	floating point	Base register ⁽²⁾ for axis + 136	RR ⁽³⁾	Maximum rate (units/second) of current move.
Maximum Setpoint ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 146	–	–
Minimum Setpoint ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 148	–	–
Maximum Velocity ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 150	–	–
Minimum Velocity ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 152	–	–
Deceleration ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 168	–	–
Acceleration ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 170	–	–
Fault Torque ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 172	–	–
Home Torque ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 174	–	–
Max Torque ⁽⁴⁾	Read/write	floating point	Base register ⁽²⁾ for axis + 176	–	–

(1) All registers are 32 bit. Byte order = CD AB. Bit format for DIN = BIN(CD) BIN(AB).

(2) See [Modbus Base Register](#) tables on page 47.

(3) Bisync is an optional protocol for automated valve positioning only.

(4) It is recommended that changes to this parameter be made through the controller GUI.

Homing Command - Modbus

The homing command can be issued for each axis at:

Axis address + 34 at bit 5 (1 = execute homing)

Axis Home Flag - Modbus

The home flag status can be read at:

Axis address + 6 at bit 3 (1 = homed; 0 = not homed)

Axis Home Switch - Modbus

The home switch status can be read at:

Axis address + 6 at bit 0

I/O Interface

I/O Interface Assignments

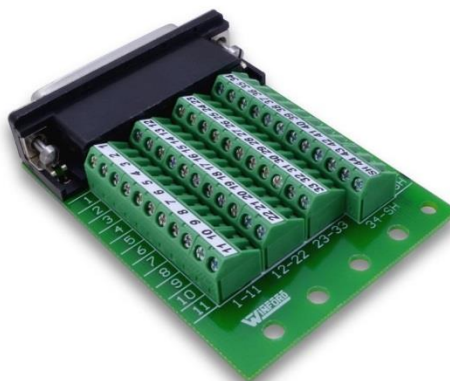
The I/O interface assignments are listed in the following table.

I/O Type	CH	Function	
		MAC	DMC
Digital Input	0	Robot Homing Switch, or Molly Axis 1 Home S/W	Molly Axis #1 Home S/W
	1	Molly Axis 3 Home S/W	Molly Axis #2 Home S/W
	2	Interlock Input 1	
	3	Molly Axis 4 Home S/W or ETO	ETO
	4	2nd Setpoint	
	5	LL Door switch	N/A
	6	Molly Axis 6 Home S/W or ETO	ETO
	7	Molly Axis 8 Home S/W or ETO	ETO
Digital Output	0	1 st stage rotation axis home pulse output	Output home pulse if Axis 1 is stage rotation
	1	2 nd stage rotation axis home pulse output	Output home pulse if Axis 2 is stage rotation
	2	ETO	ETO
	3	ETO	ETO
	4	ETO	ETO
	5	ETO	ETO
	6	Brake 1, LL Elevator	ETO
	7	Brake 2, QCM Arm	Used by Brake if Axis 1 is QCM Arm
Analog Input	0	Analog cmd- Molly Axis 1	Analog cmd- Molly Axis 1
	1	Analog cmd- Molly Axis 3	Analog cmd- Molly Axis 2
	2	Analog cmd- Molly Axis 2	ETO
	3	Analog cmd- Molly Axis 4	ETO
Analog Output	0	Analog pos-Molly axis 1	Analog cmd- Molly Axis 1
	1	Analog pos-Molly axis 3	Analog cmd- Molly Axis 2
	2	Analog pos-Molly axis 2	ETO
	3	Analog pos-Molly axis 4	ETO

Interface Breakout Board

DMC Controller

On a DMC Controller, the I/O interface is a 44-position integrated breakout board, as shown below.



I/O Interface – DMC Controller

MAC Controller

In some cases, MAC controllers include the I/O interface shown below. This is a DIN-rail-mount, 44-position breakout board. Not all MAC controllers include this option.



I/O Interface - MAC Controller (Option)

MAC Digital and Analog I/O Pinout

If the 44-position breakout board option is included with your MAC Controller, the pinout is as shown below.

Pin #	Signal Designator	Description
1	OUT1	Home pulse output (for the second stage rotation axis, if equipped)
2	OUT3	Spare
3	OUT5	Spare
4	OUT7	Linear positioner brake signal
5	IN1	Axis #2 home switch input or ETO
6	IN3	Axis #4 home switch input or ETO
7	AGND	Analog ground
8	AIN10-	Axis #1 analog command -
9	AIN11-	Axis #2 analog command -
10	AIN14-	Axis #3 analog command -
11	AIN15-	Axis #4 analog command -
12	AOUT10-	Axis #1 position indication -
13	AOUT11-	Axis #2 position indication -
14	AOUT14-	Axis #3 position indication -
15	AOUT15-	Axis #4 position indication -
16	OUT0	Home pulse output (for the first stage rotation axis, if equipped)
17	OUT2	Spare
18	OUT4	Spare
19	OUT6	Load-lock elevator motor brake
20	IN0	Axis #1 home switch input or ETO
21	IN2	Interlock Input
22	-	Not connected
23	AIN10+	Axis #1 analog command +
24	AIN11+	Axis #2 analog command +
25	AIN14+	Axis #3 analog command +
26	AIN15+	Axis #4 analog command +
27	AOUT10+	Axis #1 position indication +
28	AOUT11+	Axis #2 position indication +
29	AOUT14+	Axis #3 position indication +
30	AOUT15+	Axis #4 position indication +
31	V_SUP_IO	Input/output supply+, 24V
32	V_RTN_IO	Input/output supply return
33	IN4	Second setpoint trigger input
34	IN5	Load-lock door switch input (if equipped)
35	IN6	Axis #6 home switch input or ETO
36	IN7	Axis #8 home switch input or ETO
37	AGND	Analog ground
38	AGND	Analog ground
39	-	-

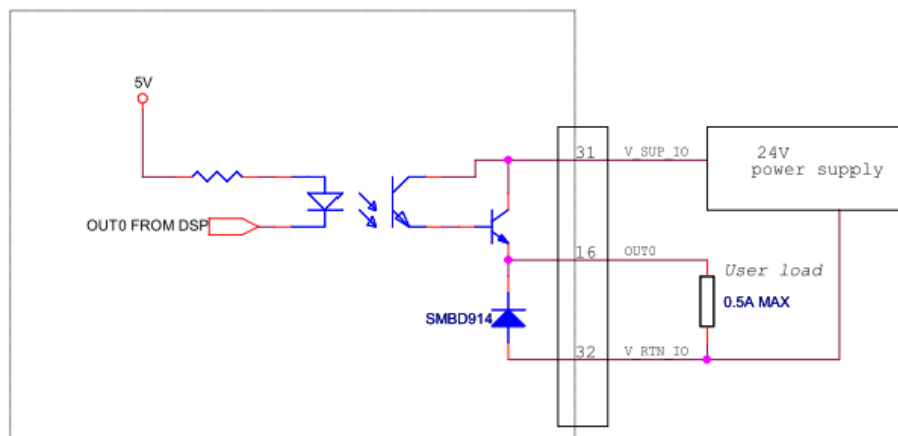
Pin #	Signal Designator	Description
40	-	-
41	-	-
42	+12V1	12V for joystick (optional, ETO)
43	12V1_RTN	+12V return for joystick (optional, ETO)
44	-12V1	-12V for joystick (optional, ETO)

DMC Digital and Analog I/O Pinout

Pin #	Signal Designator	Description
1	OUT1	Digital Output 1 (Spare or ETO)
2	OUT3	Digital Output 3 (Spare or ETO)
3	OUT5	Digital Output 5 (Spare or ETO)
4	OUT7	Digital Output 7 (Spare or ETO)
5	IN1	Digital Input 1, Axis #2 home switch input or ETO
6	IN3	Digital Input 3 (Spare or ETO)
7	N/A	-
8	N/A	-
9	N/A	-
10	AIN0-	Analog Input 0 inverted, Axis #1 analog command -
11	AOUT0+	Analog Output 0 non-inverted, Axis #1 position indication +
12	N/A	-
13	N/A	-
14	N/A	-
15	DGN	D Digital ground
16	OUT0	Digital Output 0, Home pulse output, if equipped
17	OUT2	Digital Output 2, (Spare or ETO)
18	OUT4	Digital Output 4, (Spare or ETO)
19	OUT6	Digital Output 6, (Spare or ETO)
20	IN0	Digital Input 0, Axis #2 home switch input or ETO
21	IN2	Digital Input 2, Interlock input
22	N/A	-
23	N/A	-
24	N/A	-
25	AIN0+	Analog Input 0 non-inverted, Axis #1 analog command +
26	AIN1+	Analog Input 1 non-inverted, Axis #2 analog command +
27	AOUT0-	Analog Output 0 inverted, Axis #1 position indication -
28	N/A	-
29	N/A	-
30	N/A	-
31	V_SUP_IO	Input/Output supply +, 24V

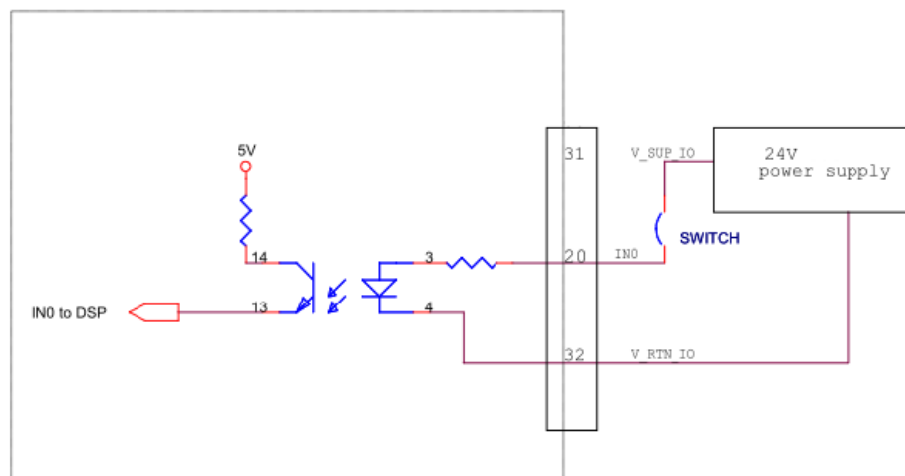
Pin #	Signal Designator	Description
32	V_RTN_IO	Input/Output supply return
33	IN4	Digital Input 4, Second setpoint trigger input
34	IN5	Digital Input 5, (Spare or ETO)
35	IN6	Digital Input 6 (Spare or ETO)
36	IN7	Digital Input 7 (Spare or ETO)
37	N/A	-
38	N/A	-
39	ANGD	Analog ground (Shield of analog command)
40	AIN1-	Analog Input 1 inverted, Axis #2 analog command -
41	AOUT1+	Analog Output 1 non-inverted, Axis #2 position indication +
42	AOUT1-	Analog Output 1 inverted, Axis #2 position indication -
43	N/A	-
44	N/A	-

Home Pulse Output Wiring



Home Pulse Output Example

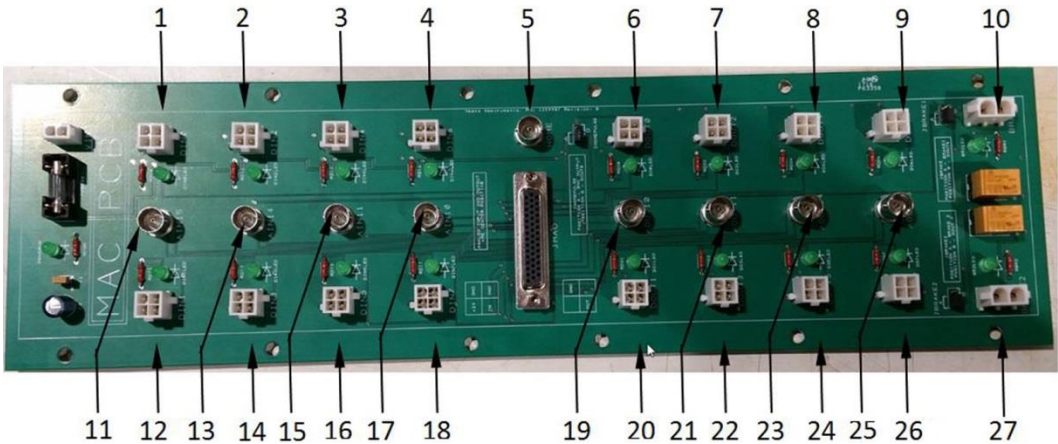
Home Switch Input Wiring



Home Switch Input Example

MAC Interface Board Connection

When installed in the GM1 electronics rack for an automated Veeco MBE system, the MAC Controller is equipped with the interface PCB shown below.



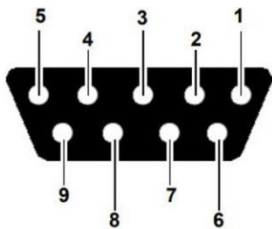
MAC Interface Board Connection

Pin	Description	Pin	Description	Pin	Description
1	DIN 0	10	Brake power output for the first axis with a brake motor	19	Analog output CH0
2	DIN 2	11	Analog input CH3	20	Home pulse output for the 2 nd stage rotation axis
3	DIN 4	12	DIN1	21	Analog output CH1
4	DIN 6	13	Analog input CH2	22	DOUT3
5	Home pulse output for the first stage rotation axis	14	DIN3	23	Analog output CH2
6	DOUT0	15	Analog input CH1	24	DOUT5
7	DOUT2	16	DIN5	25	Analog output CH3
8	DOUT4	17	Analog input CH0	26	DOUT7
9	DOUT6	18	DIN7	27	Brake power output for the second axis with a brake motor

BISYNC Protocol Converter



BISYNC Protocol Converter



Pin	RS485 Signal
1	Common
2	Transmit -
3	Receive -
4	Not used
5	Not used
6	Not used
7	Receive +
8	Transmit +
9	Not used

Pinout – Bisync Protocol Converter

MODBUS RTU Connection

On the MAC Controller the J20 connector is for MODBUS RTU via RS232 (using null modem cable to connect to the RS232 COM port on your computer). The pinout is given in the table below.

Pin #	Signal Designator	Description
1	SHIELD	Cable shield connection
2	RX232	RS-232 receive signal
3	TX232	RS-232 transmit signal
4	-	Not connected
5	DGND	Digital ground.
6	-	Not connected
7	-	Not connected
8	-	Not connected
9	-	Not connected

On the DMC Controller the MODBUS RTU connection is on the J13 - COM1 Port connector.



Modular Jack 4P4C

Connector type: Modular jack – 4P4C

Mating connector type: RJ11 plug for 4P4C

Pin	Name	Description
1	RX232	RS-232 receive signal for communication port 1 (COM1)
2	TX232	RS-232 transmit signal for communication port 1 (COM1)
3	DGND	Digital ground.
4	SHIELD	Cable shield connection

Motor Torque Table



IMPORTANT — Check the model of motor installed before using the table below. Models are in the following format: **AKM##X**.

Values in green are for **motors currently used** for an axis of motion.

Values in grey are for **motors formerly used** for an axis of motion.

	Home Torque			Fault Torque		
AVP	oz-in	% Motor		oz-in	% Motor	
		AKM21C	AKM23D		AKM21C	AKM23D
AS-500 IV	20	9.5	3.6	70	33.4	12.7
AS-500V	45	21.5	8.2	120	57.3	21.8
As-2/5/6/7L IV	20	9.5	3.6	70	33.4	12.7
As-2/5/7L V	45	21.5	8.2	120	57.3	21.8
As-10/15L	95	45.3	17.3	150	71.6	27.3
Cd-200IV	12	5.7	2.2	25	11.9	4.5
Cd-200V	12	5.7	2.2	25	11.9	4.5
Cd-2000	12	5.7	2.2	50	23.9	9.1
CdTe-200IV	12	5.7	2.2	25	11.9	4.5
CdTe-200V	12	5.7	2.2	25	11.9	4.5
CdTe-2000	12	5.7	2.2	50	23.9	9.1
Hg IV	45	21.5	8.2	60	28.6	10.9
Hg V	45	21.5	8.2	60	28.6	10.9
Mg-200IV	12	5.7	2.2	25	11.9	4.5
Mg-200V	12	5.7	2.2	25	11.9	4.5
Mg-2000	12	5.7	2.2	50	23.9	9.1
P-500	30	14.3	5.5	125	59.6	22.7
P-5000	30	14.3	5.5	125	59.6	22.7
S-500	20	9.5	3.6	70	33.4	12.7
S-2000	20	9.5	3.6	70	33.4	12.7
Se-500IV	20	9.5	3.6	70	33.4	12.7
Se-500V	45	21.5	8.2	120	57.3	21.8
Se-1200	45	21.5	8.2	120	57.3	21.8
Sb-200IV	12	5.7	2.2	25	11.9	4.5
Sb-200V	12	5.7	2.2	25	11.9	4.5
Sb-2000	12	5.7	2.2	25	11.9	4.5
Te-200IV	12	5.7	2.2	25	11.9	4.5
Te-200V	12	5.7	2.2	25	11.9	4.5
Te-2000	12	5.7	2.2	25	11.9	4.5
Zn-200IV	12	5.7	2.2	25	11.9	4.5
Zn-200V	12	5.7	2.2	25	11.9	4.5
Zn-2000	12	5.7	2.2	25	11.9	4.5
100V-LT	12	5.7	2.2	25	11.9	4.5
Linear	oz-in	% Motor		oz-in	% Motor	
		AKM23D			AKM23D	
Elevator NOT GEN10	90	16.4		292	53.1	
Elevator-GEN10	40	7.3		292	53.1	
Linear Positioner	55	10.0		292	53.1	
Growth Stage	oz-in	% Motor		oz-in	% Motor	
		AKM23D			AKM23D	
CAR	N/A	5.0		400	72.7	
INDEX	N/A	2.0		200	36.3	

	Home Torque			Fault Torque				
Robot	oz-in	% Motor			oz-in	% Motor		
		AKM43H	AKM23D	AKM32E		AKM43H	AKM23D	AKM32E
Robot-GEN200/2000	137	6.9	-	-	273	13.8	-	-
Robot-GEN20A	102	5.1	18.5	-	273	13.8	49.6	-
Robot-GEN10	43	-	-	5.1	115	-	-	13.6
Flipper	oz-in	% Motor			oz-in	% Motor		
		AKM23D				AKM23D		
Flipper-Clamping	28	5.1			220	40.0		
Flipper-Rotation	N/A	3.0			220	40.0		